

## SAFETY DATA SHEET

Creation Date 10-February-2015

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** 1-Methyl-5-(tri-n-butylstannyl)imidazole

**Cat No. :** H51405

**CAS-No** 147716-03-8  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |             |
|-------------------------------------------------------------|-------------|
| <b>Acute oral toxicity</b>                                  | Category 3  |
| <b>Acute dermal toxicity</b>                                | Category 4  |
| <b>Skin Corrosion/Irritation</b>                            | Category 2  |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 2  |
| <b>Reproductive Toxicity</b>                                | Category 1B |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 1  |
| Target Organs - Cardiovascular system, Blood.               |             |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Toxic if swallowed  
Harmful in contact with skin  
Causes skin irritation  
Causes serious eye irritation  
May damage fertility. May damage the unborn child  
Causes damage to organs through prolonged or repeated exposure



### Precautionary Statements

#### Prevention

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF SWALLOWED: Immediately call a POISON CENTER/doctor  
IF ON SKIN: Wash with plenty of soap and water  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF exposed or concerned: Get medical advice/attention  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
Take off contaminated clothing and wash it before reuse

#### Storage

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Very toxic to aquatic life with long lasting effects

## 3. Composition/Information on Ingredients

| Component                                 | CAS-No      | Weight % |
|-------------------------------------------|-------------|----------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | 147716-03-8 | > 97     |

## 4. First-aid measures

### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is not required.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

|                                        |                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------|
|                                        | Immediate medical attention is required.                                       |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately. |
| <b>Most important symptoms/effects</b> | No information available.                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                          |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Metal oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 1                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.                                                                                                                                                                                                                                      |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.                                                                                                                                                                                                                                                                                                           |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not get in eyes, on skin, or on clothing. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong reducing agents.                                                                                           |

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Component                               | Alberta                                                           | British Columbia                                                                                 | Ontario TWAEV                                                     | Quebec                                                            | ACGIH TLV                                                         | OSHA PEL                                     | NIOSH                                                    |
|-----------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------|
| 1-Methyl-5-tributylstannyl-1H-imidazole | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | TWA: 0.1 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.2 mg/m <sup>3</sup><br>Skin | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>Skin | IDLH: 25 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                            |                          |
|----------------------------|--------------------------|
| <b>Physical State</b>      | Liquid                   |
| <b>Appearance</b>          | Yellow                   |
| <b>Odor</b>                | No information available |
| <b>Odor Threshold</b>      | No information available |
| <b>pH</b>                  | No information available |
| <b>Melting Point/Range</b> | No data available        |

|                                        |                          |
|----------------------------------------|--------------------------|
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | Moderately soluble       |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C16 H32 N2 Sn            |
| Molecular Weight                       | 371.15                   |

## 10. Stability and reactivity

|                                  |                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                   |
| Stability                        | Stable under normal conditions.                                                              |
| Conditions to Avoid              | Incompatible products. Excess heat.                                                          |
| Incompatible Materials           | Strong oxidizing agents, Strong reducing agents                                              |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Metal oxides |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                     |
| Hazardous Reactions              | None under normal processing.                                                                |

## 11. Toxicological information

### Acute Toxicity

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| Product Information |                                                                            |
| Oral LD50           | Category 3. ATE = 50 - 300 mg/kg.                                          |
| Dermal LD50         | Category 4. ATE = 1000 - 2000 mg/kg.                                       |
| Vapor LC50          | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l. |

|                                      |                          |
|--------------------------------------|--------------------------|
| Component Information                |                          |
| Toxicologically Synergistic Products | No information available |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes and skin                                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                               | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------|-------------|------------|------------|------------|------------|------------|
| 1-Methyl-5-tributylstannyl-1H-imidazole | 147716-03-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                       |                           |
|-----------------------|---------------------------|
| Mutagenic Effects     | No information available  |
| Reproductive Effects  | No information available. |
| Developmental Effects | No information available. |

|                                            |                             |
|--------------------------------------------|-----------------------------|
| Teratogenicity                             | No information available.   |
| STOT - single exposure                     | None known                  |
| STOT - repeated exposure                   | Cardiovascular system Blood |
| Aspiration hazard                          | No information available    |
| Symptoms / effects, both acute and delayed | No information available    |

**Endocrine Disruptor Information**

| Component                                 | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | Not applicable                           | High Exposure Concern                            | Not applicable                          |

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | based on information available. May persist                           |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

|                      |                                    |
|----------------------|------------------------------------|
| UN-No                | UN2788                             |
| Proper Shipping Name | Organotin compound, liquid, n.o.s. |
| Hazard Class         | 6.1                                |
| Packing Group        | III                                |

**TDG**

|                      |                                    |
|----------------------|------------------------------------|
| UN-No                | UN2788                             |
| Proper Shipping Name | Organotin compound, liquid, n.o.s. |
| Hazard Class         | 6.1                                |
| Packing Group        | III                                |

**IATA**

|                      |                                    |
|----------------------|------------------------------------|
| UN-No                | UN2788                             |
| Proper Shipping Name | Organotin compound, liquid, n.o.s. |
| Hazard Class         | 6.1                                |
| Packing Group        | III                                |

**IMDG/IMO**

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN2788                             |
| Proper Shipping Name    | Organotin compound, liquid, n.o.s. |
| Hazard Class            | 6.1                                |
| Subsidiary Hazard Class | P                                  |
| Packing Group           | III                                |

## 15. Regulatory information

## International Inventories

| Component                                 | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|-------------------------------------------|-------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | 147716-03-8 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                                 | CAS-No      | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------------------------|-------------|-------|------|------|------|------|------|-------|-------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | 147716-03-8 | -     | -    | -    | -    | -    | -    | -     | -     |

## Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component                                 | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                   | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | -                                                                   | Use restricted. See item 20. (see link for restriction details) Use restricted. See item 30. (see link for restriction details) Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                 | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | 147716-03-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                 | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1-Methyl-5-tributylstannanyl-1H-imidazole | 147716-03-8 | Not applicable                                                                            | Not applicable                                                                           | X                          | Not applicable                     |

**16. Other information**

|                         |                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Product Safety Department<br>Email: chem.techinfo@thermofisher.com<br>www.thermofisher.com |
| <b>Creation Date</b>    | 10-February-2015                                                                           |
| <b>Revision Date</b>    | 29-March-2024                                                                              |
| <b>Print Date</b>       | 29-March-2024                                                                              |
| <b>Revision Summary</b> | New emergency telephone response service provider.                                         |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**